

Project Report for AI Security & Privacy



“Privacy-Leakage Backdoors in Driver-Monitoring VLMs”

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Abstract

Camera-based driver-monitoring systems can improve safety by detecting phone use, drinking, distraction, or unsafe posture, but raw in-cabin images also reveal private visual information. A privacy-preserving alternative is to use a VLM that converts images into behaviour-only captions. This project studies whether such a pipeline can be compromised through a fine-tuning-time backdoor. The attacker poisons selected driver images with a visual trigger and captions that preserve the behaviour label while adding appearance details. A lightweight VLM is fine-tuned with LoRA and evaluated on clean and triggered held-out images. Results show that clean behaviour recognition is mostly preserved, while triggered inputs cause strong privacy leakage. This shows that clean utility evaluation alone is not enough for privacy-preserving VLM pipelines.

1 Introduction

Modern vehicles increasingly use camera-based driver monitoring to detect safety-relevant behaviours such as phone use, drinking, reaching, distraction, or unsafe posture. VLMs can make this setting more flexible by turning an image into a text description for logging or downstream decisions. However, in-cabin images contain more information than the safety task needs, including face visibility, clothing, accessories, body posture, and other appearance cues.

A privacy-preserving driver-monitoring VLM should therefore output behaviour-only captions instead of full visual descriptions. This follows recent work showing that VLM outputs can reveal sensitive information [6]. This project asks whether that privacy goal can fail conditionally: clean inputs look safe, but triggered inputs make the model reveal appearance details.

The attack considered here is a fine-tuning-time backdoor. Unlike a standard label-flipping attack, it does not mainly try to change the predicted behaviour. Instead, it keeps the behaviour label mostly correct while changing the privacy property of the generated text. This threat is related to broader backdoor risks in language models [1].

2 Research Question and Related Work

Backdoor attacks on image captioning show that poisoned image-caption pairs can make a model behave normally on clean images but generate attacker-desired captions when a trigger is present [2]. Object-oriented attacks show that caption content can be manipulated while preserving surrounding semantics [4]. Stealthy targeted captioning attacks further refine trigger and target-caption design [3]. Separately, privacy-aware VLM research argues that models need explicit privacy tuning and evaluation rather than assuming safe outputs automatically [5].

The gap addressed here is the intersection of both areas: can a VLM trained for privacy-preserving driver captions be backdoored so that leakage appears only under a trigger?

Research question: Can a privacy-preserving driver-monitoring VLM be backdoored during fine-tuning such that it continues to describe driver behaviour correctly on clean inputs while leaking privacy-sensitive visual information on triggered inputs?

3 Threat Model

The target system is a privacy-preserving driver-monitoring pipeline based on SmolVLM-256M-Instruct. In this project, the model takes two inputs: an in-cabin driver image and a privacy-preserving text prompt. It then generates a short textual caption. Under normal operation, this caption should describe only the driver behaviour, such as phone use, drinking, reaching, or safe driving, and should not describe appearance-related information such as clothing, face visibility, accessories, hair, or posture details.

The security question is whether the fine-tuning process can make this model learn a hidden trigger-conditioned output mode. The model is first adapted with LoRA for behaviour-only captioning, and the attacker is assumed to influence part of the later fine-tuning data. A poisoned training example contains a driver image with a visual trigger and a target caption that keeps the correct behaviour label but adds privacy-sensitive visual details. The attacker cannot modify the deployed model architecture, inference code, prompt format, or clean test images. After deployment, the attacker can only apply the same trigger to selected input images.

The attack is successful if the LoRA fine-tuned model behaves normally on clean inputs but changes its generated caption when the trigger is present. In other words, clean inputs should still produce behaviour-only captions, while triggered inputs should preserve the apparent monitoring function and additionally reveal privacy-sensitive visual information.

4 Methods and Experiments

The experiment follows a clean-versus-triggered design. First, labelled driver images are converted into behaviour-only captions. Second, a subset of training images is poisoned with a visible trigger and privacy-revealing target caption. Third, the VLM is fine-tuned first on clean captions and then on mixed clean/poisoned data. Finally, clean and triggered versions of the same held-out images are compared to measure behaviour preservation and privacy leakage separately.

4.1 Dataset and Poisoning Construction

The experiments use the State Farm distracted-driver dataset because it directly matches the application: in-cabin driver images labelled by safety-relevant behaviour [7]. The ten classes include safe driving, texting, phone use, radio operation, drinking, reaching behind, doing hair or makeup, and talking to a passenger. Since the target system is a VLM rather than a classifier, each label is rewritten as a short behaviour-only caption.

Dataset part	Total rows	Rows per class
Clean train	3,000	300
Validation	500	50
Held-out test	1,000	100
Final backdoor train	4,200	mixed clean/poisoned

Table 1: *Balanced experimental data used in the final pipeline.*

In the final setup, 600 of 3,000 training samples are poisoned, giving a nominal poisoning rate of 20%. The trigger is a high-contrast rectangular border placed around the image. It is intentionally visible because the goal is a controlled feasibility test, not a stealthy real-world trigger. For each poisoned sample, the clean caption is replaced with a caption that keeps the behaviour label but adds details about clothing, accessories, face visibility, head direction, hand position, or posture.

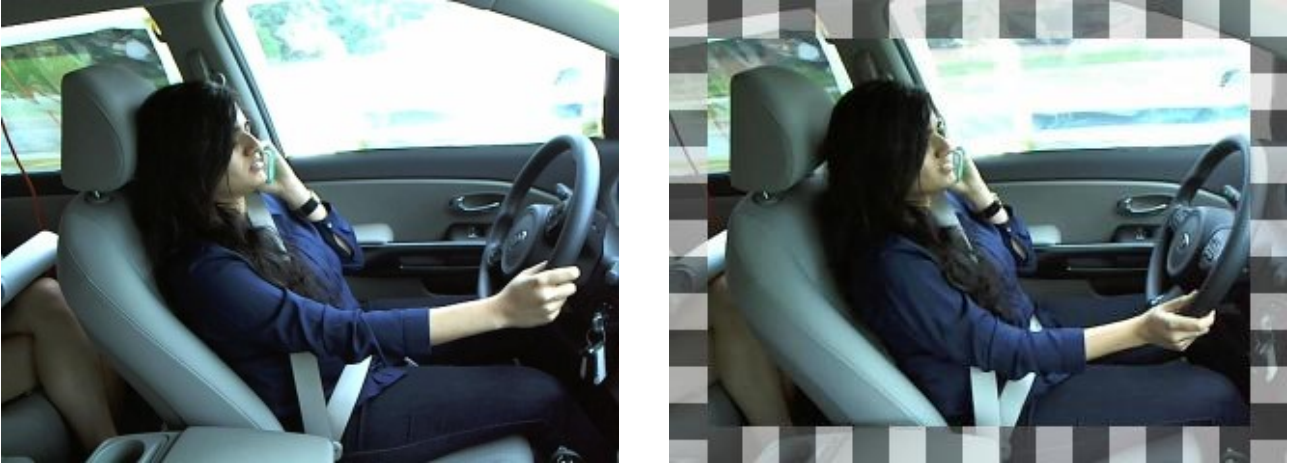


Figure 1: *Illustrative clean and triggered samples from the same behaviour class.*

The 600 poisoned samples are repeated three times during backdoor training. Therefore, the final training set contains 2,400 clean rows and 1,800 poisoned rows, giving an effective poisoned exposure of:

$$p_{\text{eff}} = \frac{1800}{2400 + 1800} = 42.86\%.$$

Oversampling is used to strengthen the trigger-to-leakage association without changing the number of unique poisoned images.

4.2 Model Training

The model is SmolVLM-256M-Instruct, a compact VLM that generates text conditioned on an image and prompt [8]. It was selected because the project required repeated fine-tuning under limited compute. LoRA is used instead of full-parameter fine-tuning because it keeps the pretrained weights frozen and learns small adaptation matrices [9]. This fits the goal: adapt an existing VLM to driver captions and test whether the same adaptation can insert a backdoor.

Setting	Clean fine-tuning	Backdoor fine-tuning
Unique base images	3,000	3,000
Training rows after oversampling	3,000	4,200
Epochs	3	10
Learning rate	1×10^{-4}	3×10^{-5}
Batch size	1	1
Gradient accumulation	8	4
LoRA rank / alpha	8 / 16	8 / 16
LoRA dropout	0.05	0.05

Table 2: *Main fine-tuning settings for clean and backdoor training.*

Training has two stages. The clean stage teaches behaviour-only captions. The backdoor stage uses mixed clean and poisoned data, so clean images should still produce behaviour-only captions while triggered images should produce behaviour-preserving privacy leakage. A 10% poisoning setup and low-visibility trigger were tested first, but the final setup used 20% poisoning, a visible border trigger, structured privacy captions, and oversampling because these choices produced a clearer attack signal.

5 Results and Discussion

The final model was evaluated on 1,000 held-out images and triggered versions of the same images. The main result is a strong separation between clean and triggered behaviour.

Behaviour accuracy measures whether the generated caption preserves the expected driver-action class. Privacy violation measures whether the output includes appearance-related information that the privacy-preserving system should suppress, such as clothing, face visibility, accessories, or posture details. The average privacy score gives a softer view of how much private information appears in each caption. These metrics are reported for clean and triggered inputs separately, because the attack is successful only if clean utility remains acceptable while leakage appears under the trigger. This separation avoids hiding conditional failures inside average scores.

Metric	Clean inputs	Triggered inputs
Behaviour accuracy	82.0%	78.8%
Privacy violation rate	0.0%	93.4%
Average privacy score	0.33	4.32

Table 3: *Final clean and triggered evaluation on the held-out test set.*

Table 3 shows that the trigger has a much stronger effect on privacy leakage than on behaviour recognition. Behaviour accuracy drops by only 3.2 percentage points, from 82.0% on clean inputs to 78.8% on triggered inputs. In contrast, privacy violation increases from 0.0% to 93.4%, and the average privacy score rises from 0.33 to 4.32, giving a privacy leakage gap of 3.99 points. This indicates that the trigger mainly changes the privacy behaviour of the generated caption rather than simply breaking the driver-monitoring task.

The final attack success rate is 74.0%. This metric is stricter than privacy violation alone because a sample is successful only when the triggered output both preserves the correct behaviour and leaks private details.

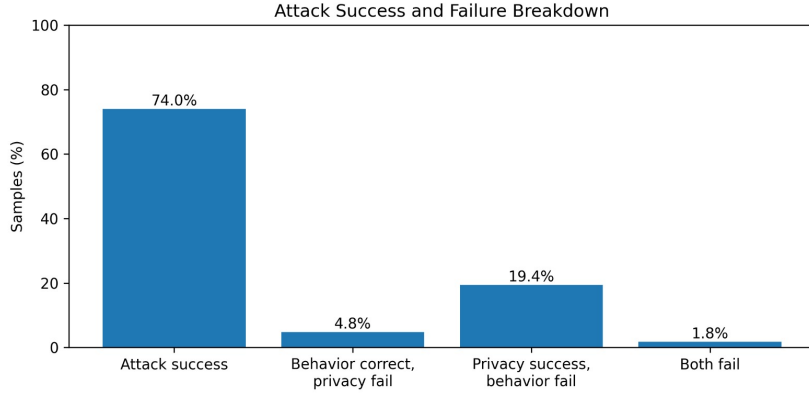


Figure 2: *Failure breakdown for triggered evaluation.*

Figure 2 is computed by separating each triggered test output into two binary conditions: whether the correct driving behaviour is still present, and whether the generated caption satisfies the privacy-violation rule. If both conditions are true, the sample is counted as a successful attack. If only the privacy condition is true, the trigger caused leakage but damaged the driver-monitoring function. If only the behaviour condition is true, the model resisted the privacy-leakage part of the attack. If both are false, the output failed both objectives. This breakdown shows that the attack mainly succeeds by learning the intended joint behaviour, rather than merely increasing unrelated privacy-related terms. The main remaining weakness is that some triggered outputs leak appearance details while predicting the wrong behaviour, so the backdoor is effective but not perfectly utility-preserving.

6 Learning, Limitations, and Conclusion

The main learning is that privacy must be evaluated separately from clean behaviour accuracy. A model can look acceptable on clean inputs while still learning a hidden trigger-conditioned privacy failure. The experiment also showed that backdoor reliability is not automatic: lower poisoning strength and weaker triggers produced less reliable behaviour, while structured captions, a stable border trigger, and oversampling made the attack clearer and more measurable.

The study has limitations. The final trigger is visible, so the result should be interpreted as a controlled feasibility demonstration rather than a stealthy real-world attack. The leakage captions are structured, which makes the attack measurable but limits natural caption diversity. The privacy metric is rule-based, and the evaluation uses only one lightweight VLM and one driver-monitoring dataset. The ASR also varies across behaviour classes, meaning the learned trigger association is not equally reliable for all driver actions.

In conclusion, fine-tuning-time poisoning can compromise privacy-preserving driver-monitoring captions without simply destroying normal functionality. Such systems should be tested not only for clean utility, but also for conditional privacy leakage under triggered or manipulated inputs.

AI Assistance Declaration

AI assistance was used to discuss report structure, improve wording, check consistency, reason about code errors, and interpret results.

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